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EVERY INTERMEDIATE LOGIC IS AXIOMATIZABLE BY FORMULAS MINIMAL IN CLASSICAL LOGIC

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Abstract. A formula is minimal in a logic L if it belongs to L and no strictly more general formula, with respect to the substitution preorder, belongs to L. Komori and Kashima asked whether intuitionistic and classical logic are the only implicational intermediate logics axiomatizable by formulas minimal in classical logic. Nakamura and Matsuda refuted this with an ad hoc axiom of Gödel–Dummett logic and asked which logics are so axiomatizable. We show that, surprisingly, all of them are: for every classical tautology Φ we construct, in linear time, a formula M minimal in classical logic such that $\mathbf{IL} + M = \mathbf{IL} + \Phi$, and M can be taken of order at most 4, which is optimal. The same holds for intermediate logics in the language $\{\bot, \wedge, \vee, \to\}$. Thus being axiomatizable by minimal formulas is not a property of a logic but of the presentation of its axioms. Under the Curry–Howard correspondence, a formula is minimal in L exactly when it is the principal type of every proof of it in L; in contrast to Hindley's theorem that every inhabited type is the principal type of some term, our result says that every intermediate logic is axiomatized by types that are the principal types of all their classical inhabitants.

§1. Introduction. A formula β is more general than a formula α, written $\beta \leqslant_{\mathrm{sub}} \alpha$, if α is a substitution instance of β, and strictly more general, written $\beta <_{\mathrm{sub}} \alpha$, if moreover β is not a substitution instance of α. A formula α is minimal in a logic L if $\alpha \in L$ and no formula strictly more general than α belongs to L. For instance, the axioms $K = a \to b \to a$, $S = (a \to b \to c) \to (a \to b) \to a \to c$ and Peirce's law $P = ((a \to b) \to a) \to a$ are minimal in classical logic **CL**, whereas $(a \to b) \to a \to b$ is not, since it is an instance of $a \to a$.

The substitution preorder was introduced by Komori [Kom87] in the study of **BCK**-algebras and the λ-calculus, and minimal formulas have a natural reading under the Curry–Howard correspondence. Hirokawa showed that the formulas minimal in **BCK** are exactly the principal type-schemes of **BCK**-λ-terms in $\beta\eta$-normal form [Hir92, Hir93], and a formula minimal in **BCK** has a unique $\beta\eta$-normal proof [Tat06]; see also [Hir91, KH93]. Beyond **BCK** uniqueness fails: Tatsuta [Tat93] gave minimal formulas of depth 3 with several $\beta\eta$-normal proofs in NJ and in NK.

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[Plot: Check the exact statements and attributions in [Hir92], [Hir93], [Tat93], [Tat06]; Wroński's proof is mentioned only in secondary sources and is not cited until a written source is found.]

Since minimal formulas are the most general members of a logic, it is natural to ask which logics can be axiomatized by them. Komori and Kashima [Kom05, Kas05] asked whether intuitionistic logic **IL** and **CL** are the only implicational intermediate logics axiomatizable by formulas minimal in **CL**; Komori's original conjecture is stronger and states that over **BCK** the only such logics are **BCK**, **IL** and **CL** [Kom05, Kom07]. The expected answer was positive. Kashima gave a partial positive solution by means of a sequent calculus [Kas05], and a computer search by Nakamura and Matsuda found no small counterexample [NM21, Remark 3.4].

[Plot: State precisely which class Kashima's partial positive solution covers and check that it is consistent with Theorem 1.1; also check [Kas05b].]

Nakamura and Matsuda [NM21] nevertheless found a formula G' minimal in **CL** with $\mathbf{IL}{+}G' = \mathbf{LC}$, Gödel–Dummett logic. Their formula was obtained by an ad hoc modification of Dummett's axiom $((a \to b) \to c) \to ((b \to a) \to c) \to c$, which is not itself minimal in **CL**, and its minimality was checked case by case. They asked how many intermediate logics are axiomatizable by formulas minimal in **CL** and whether this class can be characterized [NM21, Open Problem 1], and whether such an axiom can be found among formulas of order at most 4 [NM21, Open Problem 2]. In the language with $\bot$, minimal axioms were known only for Jankov's logic **KC** [NM21, Remark 1.1].

We show that minimality in **CL** imposes no restriction at all on axiomatizability.

Theorem 1.1. *For every implicational formula $\Phi \in \mathbf{CL}$ there is an implicational formula $\widehat{M}_\Phi$ minimal in* **CL** *such that*

$$\mathbf{IL} + \widehat{M}_\Phi = \mathbf{IL} + \Phi \qquad \textit{and} \qquad \mathrm{Ord}(\widehat{M}_\Phi) \leq 4.$$

Moreover, $\widehat{M}_\Phi$ is computable from Φ in linear time, and there is a substitution σ fixing the variables of Φ such that $\mathbf{IL} \vdash \Phi \to \widehat{M}_\Phi$, $\mathbf{IL} \vdash \sigma(\widehat{M}_\Phi) \to \Phi$ and $\mathbf{IL} \vdash \Phi \to \sigma(\widehat{M}_\Phi)$.

Applying Theorem 1.1 to each axiom of an arbitrary axiomatization gives the following.

Corollary 1.2. *Every implicational intermediate logic L is of the form $L = \mathbf{IL} + \Gamma$ for a set Γ of formulas of order at most 4 minimal in* **CL**. *If $L \neq \mathbf{IL}$, the bound 4 cannot be lowered.*

The optimality follows from the fact that every formula of **CL** of order at most 3 belongs to **IL** (Proposition 4.5). Thus the Komori–Kashima problem has a negative answer in the strongest possible sense, Open Problem 1 of [NM21] is answered by "all of them", and Open Problem 2 has an affirmative answer.

The base logic can be brought under the same criterion. Since $\mathbf{IL} = \{K, S\}^*$ and K, S are minimal in **CL**, every implicational intermediate logic is the closure under modus ponens and substitution of a set of formulas minimal in **CL**, all of order at most 4; no base logic needs to be presupposed. The same applies

to Komori's base **BCK**: its axioms are minimal in **CL** and **IL** = **BCK** + S, so every implicational intermediate logic is **BCK** + Γ for a set Γ of formulas minimal in **CL** (Remark 4.6). Komori's conjecture that **BCK**, **IL** and **CL** are the only such extensions of **BCK** thus fails for every logic between **IL** and **CL**; what remains open is the interval between **BCK** and **IL**.

The same holds in the full language.

THEOREM 1.3. *For every formula Φ of* **CPC** *in the language $\{\bot, \wedge, \vee, \to\}$ there is a formula M in the same language, minimal in* **CPC**, *such that* **IPC** + M = **IPC** + Φ *and* $\mathrm{Ord}(M) \leq 4$. *Consequently, every intermediate logic is axiomatizable by formulas minimal in classical logic.*

In the full language the bound 4 is not optimal in general: $p \vee (p \to q)$ is an axiom of **CPC** of order 2 minimal in **CPC**, and $(p \to q) \vee (\neg p \to r)$ is an axiom of **KC** of order 3 minimal in **CPC** (Examples 5.5 and 5.6).

Theorems 1.1 and 1.3 say that, for an intermediate logic, "being axiomatizable by formulas minimal in **CL**" is not a property of the logic. It is a property of the way an axiom is written down: every axiom can be rewritten, without changing the logic it generates, into one that cannot be generalized while remaining classically valid. Dummett's axiom is not minimal, but **LC** has minimal axioms; this is not a feature of **LC** but of every intermediate logic. In the implicational language the only invariant that survives is the order: an intermediate logic other than **IL** has a minimal axiom of order 4 but none of order 3.

The rewriting cannot be done by local simplification. Generalizing a classical tautology along the substitution preorder may leave the logic it generates, as Dummett's axiom shows: its generalization $((a \to b) \to c) \to ((b \to d) \to c) \to c$ is still in **CL** [NM21, Example 2.4]. Our construction goes the other way. It adds structure to Φ until no generalization of the result is classically valid, while keeping a substitution that recovers Φ.

The choice of **CL** as the reference logic is essential. If minimality is taken in the logic being axiomatized, the question is trivial: every $\alpha \in L$ has, up to renaming, only finitely many generalizations, so some generalization of α is minimal in L, and therefore L = **IL** + $(L \cap \mathrm{Min}(L))$ for every logic $L \supseteq$ **IL** (Remark 2.2). Our axioms satisfy a strong form of this: since $\mathrm{Min}(\mathbf{CL}) \cap L' \subseteq \mathrm{Min}(L')$ for every $L' \subseteq$ **CL**, the axiom $\widehat{M_\Phi}$ is minimal simultaneously in every logic L' with **IL** + $\Phi \subseteq L' \subseteq$ **CL**.

Under the Curry–Howard correspondence the result has a reading in terms of principal types. Present an implicational intermediate logic L by the simply typed λ-calculus with constants whose types are the axiom schemes of L; for **CL** one may take a constant of type P, the type of `call/cc`. Then a formula $\alpha \in L$ is minimal in L if and only if α is the principal type of every closed term of type α (Proposition 6.1). By Remark 2.2, every logic is trivially axiomatized by types that are principal for all their inhabitants in that logic. Theorem 1.1 gives the classical form: every intermediate logic is axiomatized by types that are principal for all their *classical* inhabitants (Corollary 6.2).

This should be compared with two facts. Hindley's theorem [Hin69] states that every inhabited simple type is the principal type of *some* closed term, so principality for one inhabitant is no restriction on a type. Principality for *all*

inhabitants is a strong restriction: in **BCK** it comes with a unique normal inhabitant [Hir92, Tat06]. In the intermediate setting uniqueness is lost [Tat93], but our result shows that principality for all classical inhabitants costs nothing at the level of logics.

Let us outline the proof. The first layer (Section 3) turns Φ into a formula M_Φ minimal in **CL** with $\mathbf{IL} + M_\Phi = \mathbf{IL} + \Phi$. By a well-known criterion (Proposition 2.3), a classical tautology is minimal if and only if no splitting of the occurrences of a variable and no abstraction of occurrences of a compound subformula yields a classical tautology. We obtain M_Φ as

$$M_\Phi = \mathsf{T}(\mathsf{Mon}(\mathcal{C}_\Phi, t_\varepsilon)),$$

where only the middle and outer constructions are new. The list $\mathcal{C}_\Phi$ of local constraints, one for each node of the syntax tree, is obtained from two known translations: renaming apart the occurrences of variables with polarity-dependent guards, and the polarized extension-variable translation of Statman, Plaisted–Greenbaum and Mints. The monitor $\mathsf{Mon}(\mathcal{C}, r)$ works for an arbitrary list of read-once formulas: each formula is guarded by its own selector, and the guarded list is required once as a premise and once as a query, so that a modification of a single occurrence is refuted by a valuation that only has to falsify one formula of the list. The tree closure $\mathsf{T}(F)$ works for an arbitrary formula F no separation of which is classically valid: it renames apart all occurrences again and ties the new variables of each variable together by a spanning tree, whose connectivity rules out every splitting; the guards then rule out every abstraction (Theorem 3.16). The resulting formula has linear size but order up to 7.

The second layer (Section 4) lowers the order: we show that the polarized Statman translation S preserves minimality in every logic that contains I and the translated formula (Theorem 4.1). Hence $\widehat{M}_\Phi = \mathsf{S}(M_\Phi)$ satisfies Theorem 1.1. In the full language, conjunction lets us merge premise and query directly, and the first layer already has order at most 4 (Section 5).

Extension variables go back to Tseitin [Tse68] and were used by Statman [Sta79] to reduce intuitionistic propositional logic to its implicational fragment (see also [Sve03]); polarized versions are due to Plaisted and Greenbaum [PG86] and Mints [Min90]. Jeřábek [Jer16] used a translation of the same kind to show that the substructural hierarchy of [CGT08] collapses over the full Lambek calculus with exchange $\mathbf{FL_e}$, but did not consider minimality. To our knowledge, Theorem 4.1 has not been observed before. Nakamura and Matsuda [NM21, Proposition 3.3] showed that every implicational intermediate logic is axiomatizable by *simple* formulas, that is, formulas in which no implication between two fresh variables can be abstracted. Minimality requires, in addition, that no variable can be split and that no repeated compound subformula can be abstracted; these are the cases handled by the ports and guards of our first layer.

[Plot: Check work on projective formulas and unification (Ghilardi) for any statement equivalent to Theorem 1.1; check the exact forms in [Sta79], [Sve03], [Min90].]

The paper is organized as follows. Section 2 fixes notation and proves the minimality criterion. Section 3 carries out the first layer: the known translations (Section 3.1), the monitor (Section 3.2), the tree closure (Section 3.3), and their

combination (Section 3.4). Section 4 reduces the order by S and proves Theorem 1.1 and Corollary 1.2. Section 5 treats the full language, and Section 6 gives the principal type reading. Section 7 lists open problems, among them Komori's original **BCK** version of the problem, to which our construction does not apply because it uses contraction.

§2. Preliminaries.

2.1. Formulas and logics. Except in Section 5, formulas are built from propositional variables by the implication $\to$, which associates to the right: $\alpha \to \beta \to \gamma$ stands for $\alpha \to (\beta \to \gamma)$. For a finite list $\Delta = (\delta_1, \dots, \delta_k)$ of formulas we write $\Delta \to \beta$ for $\delta_1 \to \cdots \to \delta_k \to \beta$. In Section 5 the language is $\{\bot, \wedge, \vee, \to\}$, and $\neg\alpha$ abbreviates $\alpha \to \bot$.

A *substitution* σ maps variables to formulas and is extended homomorphically to all formulas; in particular $\sigma(\bot) = \bot$. A *logic* is a set of formulas closed under modus ponens and substitution. For a set F of formulas, F^* is the smallest logic containing F; for a logic L we write $L + F$ for $(L \cup F)^*$ and $L + \alpha$ for $L + \{\alpha\}$. Thus the members of F are used as axiom schemes.

We use the axiom schemes

$$K = a \to b \to a, \qquad S = (a \to b \to c) \to (a \to b) \to a \to c, \qquad I = a \to a,$$

$$B = (b \to c) \to (a \to b) \to a \to c, \qquad C = (a \to b \to c) \to b \to a \to c,$$

$$P = ((a \to b) \to a) \to a.$$

Implicational intuitionistic and classical logic are $\mathbf{IL} = \{K, S\}^*$ and $\mathbf{CL} = \mathbf{IL} + P$; we also use $\mathbf{BCK} = \{B, C, K\}^*$, $\mathbf{BCI} = \{B, C, I\}^*$ and Gödel–Dummett logic [Dum59] $\mathbf{LC} = \mathbf{IL} + ((a \to b) \to c) \to ((b \to a) \to c) \to c$. In the full language, **IPC** and **CPC** denote intuitionistic and classical propositional logic. An *intermediate logic* is a logic L with $\mathbf{IL} \subseteq L \subseteq \mathbf{CL}$, or with $\mathbf{IPC} \subseteq L \subseteq \mathbf{CPC}$ in the full language. A formula belongs to **CL** (resp. **CPC**) if and only if it is true under every Boolean valuation; we write $v(\alpha) \in \{0, 1\}$ for the value of α under a valuation v, with $v(\bot) = 0$.

The *order* of a formula is defined by $\mathrm{Ord}(p) = \mathrm{Ord}(\bot) = 1$, $\mathrm{Ord}(\alpha \to \beta) = \max\{\mathrm{Ord}(\alpha)+1, \mathrm{Ord}(\beta)\}$ and $\mathrm{Ord}(\alpha \circ \beta) = \max\{\mathrm{Ord}(\alpha), \mathrm{Ord}(\beta)\}$ for $\circ \in \{\wedge, \vee\}$. The *size* $|\alpha|$ is the number of occurrences of variables and of $\bot$ in α, and $\mathrm{Var}(\alpha)$ is the set of variables of α.

2.2. The substitution preorder. We write $\beta \leqslant_{\mathrm{sub}} \alpha$ if $\alpha = \sigma\beta$ for some substitution σ; then β is *more general* than α. We write $\beta <_{\mathrm{sub}} \alpha$ if $\beta \leqslant_{\mathrm{sub}} \alpha$ but not $\alpha \leqslant_{\mathrm{sub}} \beta$, and say that β is *strictly more general* than α. The relation $\leqslant_{\mathrm{sub}}$ is the *substitution preorder*. Since a substitution never decreases the number of occurrences of connectives and of $\bot$, we have $\alpha \leqslant_{\mathrm{sub}} \beta \leqslant_{\mathrm{sub}} \alpha$ if and only if β is obtained from α by a *renaming*, that is, by a substitution that maps $\mathrm{Var}(\alpha)$ injectively to variables.

Definition 2.1. A formula α is *minimal in* a logic L if $\alpha \in L$ and there is no $\beta \in L$ with $\beta <_{\mathrm{sub}} \alpha$. We write $\mathrm{Min}(L)$ for the set of formulas minimal in L.

If $L \subseteq L'$, then $\mathrm{Min}(L') \cap L \subseteq \mathrm{Min}(L)$.

REMARK 2.2. Up to renaming, a formula α has only finitely many formulas $\beta \leqslant_{\mathrm{sub}} \alpha$. Hence for every $\alpha \in L$ there is $\beta \in \mathrm{Min}(L)$ with $\beta \leqslant_{\mathrm{sub}} \alpha$, and $\alpha \in L_0 + \beta$ for every logic L_0. Consequently $L = L_0 + (L \cap \mathrm{Min}(L))$ for every logic $L_0 \subseteq L$: every logic is axiomatizable by formulas minimal in itself. The content of Theorems 1.1 and 1.3 is that the axioms can be chosen minimal in **CL** (resp. **CPC**), and hence minimal in every logic between the given one and **CL** (resp. **CPC**) at the same time.

2.3. Occurrences and a criterion for minimality. We identify an *occurrence* in a formula α with its position in the syntax tree, that is, with a word over $\{0,1\}$: the empty word ε is the root, and if the subformula at u is $\gamma \circ \delta$ with $\circ \in \{\wedge, \vee, \to\}$, then γ is at $u0$ and δ at $u1$. We write α_u for the subformula at u. An occurrence u is *strictly below* v if v is a proper prefix of u. An occurrence u is a *leaf* if α_u is a variable or $\bot$, and a *non-leaf occurrence* otherwise. Distinct occurrences of the same formula are never strictly below one another. We write $\mathrm{Occ}_\alpha(p)$ for the set of occurrences of a variable p in α.

Each occurrence has a *polarity*, *positive* or *negative*: the root is positive; if α_u is an implication, then $u1$ has the polarity of u and $u0$ the opposite one; if α_u is a conjunction or a disjunction, then $u0$ and $u1$ have the polarity of u. If $\alpha = \Delta \to \beta$, every occurrence in a member of Δ has, in α, the opposite of its polarity in that member.

Let Q be a nonempty set of occurrences of a single formula δ in α, and let z be a variable not occurring in α. We write $\alpha[Q{:=}z]$ for the result of replacing the subformula at each element of Q by z. We call it a *splitting* of α if δ is a variable and $Q \neq \mathrm{Occ}_\alpha(\delta)$, and an *abstraction* of α if δ is not a variable (that is, δ is $\bot$ or a compound formula). A splitting with $|Q| = 1$ is a *separation*. The splittings and abstractions of α are its *elementary generalizations*; we write $\mathcal{G}(\alpha)$ for the set of them.

PROPOSITION 2.3. *Let L be a logic and $\alpha \in L$. Then α is minimal in L if and only if $\mathcal{G}(\alpha) \cap L = \emptyset$.*

For implicational formulas this is essentially [NM21, Props. 3.1 and 3.2]; we include a proof.

PROOF. *"Only if".* Let $\beta = \alpha[Q{:=}z] \in \mathcal{G}(\alpha)$, where Q is a set of occurrences of δ. Substituting δ for z in β yields α, so $\beta \leqslant_{\mathrm{sub}} \alpha$. We show that $\alpha \not\leqslant_{\mathrm{sub}} \beta$, so that $\beta \notin L$ by minimality. Suppose $\tau\alpha = \beta$. If δ is not a variable, then β has fewer occurrences of connectives and $\bot$ than α, which is impossible. If δ is a variable p, then τ maps $\mathrm{Var}(\alpha)$ to variables, so $\tau\alpha$ carries the single variable $\tau(p)$ at all leaves in $\mathrm{Occ}_\alpha(p)$, whereas β carries z at the elements of Q and p at the elements of $\mathrm{Occ}_\alpha(p) \setminus Q \neq \emptyset$; a contradiction.

"If". Suppose $\mathcal{G}(\alpha) \cap L = \emptyset$, and let $\gamma \in L$ and $\sigma\gamma = \alpha$. If σ maps $\mathrm{Var}(\gamma)$ injectively to variables, then $\alpha \leqslant_{\mathrm{sub}} \gamma$. Otherwise there is $q \in \mathrm{Var}(\gamma)$ such that either $\sigma(q)$ is not a variable, or $\sigma(q) = \sigma(q')$ is a variable for some $q' \in \mathrm{Var}(\gamma)$ with $q' \neq q$. Let σ' agree with σ except that $\sigma'(q) = z$ for a variable z not in α. The occurrences of q in γ give a nonempty set Q of occurrences of $\sigma(q)$ in α with $\sigma'\gamma = \alpha[Q{:=}z]$, and in the second case the occurrences of q' give occurrences of $\sigma(q)$ outside Q. Hence $\sigma'\gamma \in \mathcal{G}(\alpha) \cap L$, a contradiction. ⊣

A splitting that divides $\mathrm{Occ}_\alpha(p)$ into more than two classes is not needed: identifying all classes but one yields a splitting in the above sense, which is an instance of the former.

§3. Implicational intermediate logics. In this section we construct, for every implicational formula $\Phi \in \mathbf{CL}$, a formula M_Φ minimal in **CL** with $\mathbf{IL} + M_\Phi = \mathbf{IL} + \Phi$ (Theorem 3.17). Its order is reduced to 4 in Section 4.

By Proposition 2.3, we must make every splitting and every abstraction of the output fail classically, while keeping a substitution that recovers Φ. A falsifying valuation for a modified formula is easy to find when the modification is *local*, that is, when it affects a small read-once piece of the formula on which the rest of the formula imposes no condition. We achieve this by composing two known translations with two general constructions:

$$M_\Phi = \mathsf{T}(\mathsf{Mon}(\mathcal{C}_\Phi, t_\varepsilon)).$$

Here $\mathcal{C}_\Phi$ is the list of local constraints of Φ, obtained by renaming apart the occurrences of variables and applying the polarized extension-variable translation (Section 3.1). The monitor $\mathsf{Mon}(\mathcal{C}, r)$ turns any list of read-once formulas into a formula no separation of which is classically valid (Section 3.2), and the tree closure $\mathsf{T}(F)$ turns any such formula into a minimal one (Section 3.3). The two general constructions do not depend on Φ. We use Peirce's law

$$P = ((a \to b) \to a) \to a$$

as a running example, and we explain after each step which elementary generalizations survive it.

3.1. Two known translations. The first translation is *renaming apart* the occurrences of variables. Let α be a formula. For every leaf o of α we take a new variable p_o, called a *port*, distinct leaves receiving distinct ports, and let α° be the result of replacing each leaf o by p_o; thus α° is read-once. If a is the variable at o, the *guard* of o is

$$G_o = \begin{cases} a \to p_o, & \text{if } o \text{ is positive in } \alpha, \\ p_o \to a, & \text{if } o \text{ is negative in } \alpha. \end{cases}$$

We write $\mathcal{G}u_\alpha$ for the list of the guards of α, call a the *center* of the ports of its occurrences, and let τ_α be the substitution that maps every port to its center and fixes all other variables. The following is the usual monotonicity argument; the guards are the clauses for leaves in the structure-preserving translation of Plaisted and Greenbaum [PG86].

LEMMA 3.1. *$\mathcal{G}u_\alpha \vdash_{\mathbf{IL}} \alpha \to \alpha^\circ$, and τ_α maps every guard to a formula $a \to a$ and α° to α.*

PROOF. By induction on subformulas, $\mathcal{G}u_\alpha$ derives $\alpha_u \to {\alpha^\circ}_u$ for positive u and ${\alpha^\circ}_u \to \alpha_u$ for negative u, since the antecedent of an implication has the opposite polarity. ⊣

The second translation is the polarized form of Statman's extension-variable translation [Sta79, PG86, Min90]. Let α be an implicational formula. For each non-leaf occurrence u of α we choose a new variable x_u, called an *extension*

variable, distinct occurrences receiving distinct variables. For every occurrence u let $r_u = \alpha_u$ if u is a leaf and $r_u = x_u$ otherwise. For a non-leaf occurrence u put $b_u = r_{u0} \to r_{u1}$ and

$$E_u = \begin{cases} b_u \to x_u, & \text{if } u \text{ is positive,} \\ x_u \to b_u, & \text{if } u \text{ is negative.} \end{cases}$$

DEFINITION 3.2. Let $u_1, \dots, u_m$ be the non-leaf occurrences of α in post-order; that is, u_i precedes u_j if and only if u_i is strictly below u_j, or $u_i = w0u'$ and $u_j = w1u''$ for some words w, u', u''. The *translation* of α is

$$\mathsf{S}(\alpha) = E_{u_1} \to \cdots \to E_{u_m} \to r_\varepsilon.$$

If α is a variable, then $\mathsf{S}(\alpha) = \alpha$.

Every b_u has order 2, so every E_u has order at most 3, and

$$\mathrm{Ord}(\mathsf{S}(\alpha)) \le 4, \qquad |\mathsf{S}(\alpha)| = 3|\alpha| - 2. \tag{1}$$

Let θ_α be the substitution that maps each x_u to α_u and fixes all other variables. Then $\theta_\alpha(r_u) = \alpha_u$, and θ_α maps every E_u to $\alpha_u \to \alpha_u$.

LEMMA 3.3. *For every implicational formula α, the formulas E_u derive $\alpha \to r_\varepsilon$ in* **IL**; *hence* $\mathbf{IL} \vdash \alpha \to \mathsf{S}(\alpha)$. *Moreover* $\mathbf{IL} \vdash \theta_\alpha(\mathsf{S}(\alpha)) \to \alpha$, *and therefore* $\mathbf{IL} + \mathsf{S}(\alpha) = \mathbf{IL} + \alpha$.

PROOF. By induction on subformulas, the formulas E_u derive $\alpha_u \to r_u$ for positive u and $r_u \to \alpha_u$ for negative u. For instance, if u is positive, then from $r_{u0} \to \alpha_{u0}$ and $\alpha_{u1} \to r_{u1}$ we obtain $\alpha_u \to b_u$, and $E_u = b_u \to x_u$ gives $\alpha_u \to x_u$. The second claim holds since $\theta_\alpha(\mathsf{S}(\alpha)) = (\gamma_1 \to \gamma_1) \to \cdots \to (\gamma_m \to \gamma_m) \to \alpha$. ⊣

Every formula E_u is used exactly once in this derivation, which in fact takes place in **BCI**.

We combine the two translations. Let Φ be an implicational formula that is not a variable. For every occurrence v of Φ we write t_v for the port of v if v is a leaf, and for the extension variable of v in Φ° otherwise; these are the *node variables*. The list $\mathcal{C}_\Phi$ of *local constraints* of Φ consists of the guards of Φ and the formulas E_u of Φ°. Explicitly, writing C_v for the constraint of v, with $u = v0$ and $w = v1$,

Φ_v	v positive	v negative
a variable p	$p \to t_v$	$t_v \to p$
$\Phi_u \to \Phi_w$	$(t_u \to t_w) \to t_v$	$t_v \to (t_u \to t_w)$

There are $2|\Phi| - 1$ local constraints; each is read-once, has two or three variables, and has order at most 3. By Lemmas 3.1 and 3.3:

LEMMA 3.4. $\mathcal{C}_\Phi \vdash_{\mathbf{IL}} \Phi \to t_\varepsilon$. *The substitution $\tau_\Phi \circ \theta_{\Phi^\circ}$, which maps t_v to Φ_v and fixes the variables of Φ, maps every local constraint to a formula $\gamma \to \gamma$ and t_ε to Φ.*

EXAMPLE 3.5. The occurrences of P, their polarities and their local constraints are as follows.

v	P_v		C_v
ε	$((a \to b) \to a) \to a$	$+$	$(t_0 \to t_1) \to t_\varepsilon$
0	$(a \to b) \to a$	$-$	$t_0 \to t_{00} \to t_{01}$
00	$a \to b$	$+$	$(t_{000} \to t_{001}) \to t_{00}$
000	a	$-$	$t_{000} \to a$
001	b	$+$	$b \to t_{001}$
01	a	$-$	$t_{01} \to a$
1	a	$+$	$a \to t_1$

Each constraint says that a node variable is an upper bound (for positive nodes) or a lower bound (for negative nodes) of the subformula at its node, in terms of the node variables of the children.

The formula $\mathcal{C}_\Phi \to t_\varepsilon$, that is, $\mathsf{S}(\Phi^\circ)$ preceded by the guards of Φ, generates the same logic as Φ over **IL** by Lemma 3.4, but it need not be minimal, even if Φ is. For P, the remaining constraints yield t_ε classically without $b \to t_{001}$: if $t_\varepsilon = 0$, then $t_0 = 1$ and $t_1 = 0$, hence $a = t_{01} = t_{000} = 0$, hence $t_{00} = 1$, contradicting $t_0 \to t_{00} \to t_{01}$. Therefore the separation that replaces t_{001} in $b \to t_{001}$ by a new variable is still in **CL**. The problem is that a falsifying valuation must satisfy all the other constraints at once, and these together express the validity of Φ.

3.2. The monitor. Let $\mathcal{C} = (C_1, \dots, C_n)$ be a list of implicational formulas and r a variable. We take new variables $x_1, \dots, x_n$, called *selectors*, a new variable s, called the *switch*, and a new variable D, and put

$$H_i = x_i \to C_i, \qquad K_i = s \to x_i, \qquad Q = s \to r,$$

$$\Gamma = (H_1, K_1, \dots, H_n, K_n), \qquad \mathsf{Mon}(\mathcal{C}, r) = \Gamma \to (Q \to \Gamma \to D) \to D.$$

We call $\mathsf{Mon}(\mathcal{C}, r)$ the *monitor* of $\mathcal{C}$ and r. Thus Γ occurs twice; we call the members of the first occurrence *left* and those of the second *right*, and write ${H_i}^L, {H_i}^R$ and so on. Every occurrence in a left member has, in $\mathsf{Mon}(\mathcal{C}, r)$, the opposite polarity of the corresponding occurrence in the right member. The *data variables* are the variables of $\mathcal{C}$ and r.

The switch turns all constraints on at once, as needed to recover the original formula; with the switch off, any single constraint can be turned on by its selector, so that a countermodel for one modified constraint need not satisfy the others. The right copy of Γ is a query that detects a modification of any left member, and vice versa.

LEMMA 3.6. *Suppose that $\mathcal{C} \vdash_{\mathbf{IL}} \Phi \to r$, and that a substitution θ fixing the variables of Φ maps every C_i to a theorem of* **IL** *and r to Φ, where the selectors, s and D do not occur in Φ. Then $\mathbf{IL} \vdash \Phi \to \mathsf{Mon}(\mathcal{C}, r)$, and $\mathbf{IL} \vdash \theta'(\mathsf{Mon}(\mathcal{C}, r)) \to \Phi$ for the extension θ' of θ that maps s and the selectors to $\top = e \to e$ for a new variable e and D to Φ. Hence $\mathbf{IL} + \mathsf{Mon}(\mathcal{C}, r) = \mathbf{IL} + \Phi$.*

PROOF. Assume Φ, the members of Γ, and $Q \to \Gamma \to D$. Under the assumption s, the formulas K_i and H_i give every C_i, hence r; thus Q holds. Applying

$Q \to \Gamma \to D$ to Q and to the members of Γ again gives D. Conversely, θ' maps every member of Γ to a theorem of **IL** and $Q \to \Gamma \to D$ to $(\top \to \Phi) \to \theta'(\Gamma) \to \Phi$, which is also a theorem. ⊣

LEMMA 3.7. *Every variable of* $\mathsf{Mon}(\mathcal{C}, r)$ *other than* r *has a positive and a negative occurrence; so does* r *if it occurs in some* C_i.

PROOF. Every variable other than D and r occurs in Γ, and corresponding occurrences in the left and right copies have opposite polarities. The variable D occurs negatively inside $Q \to \Gamma \to D$ and positively at the end. ⊣

A formula is *read-once* if no variable occurs in it twice.

LEMMA 3.8. *Let* γ *be a read-once formula without* $\bot$*, built by* $\wedge$*,* $\vee$ *and* $\to$*, and let* u *be a leaf of* γ*. Then there is a valuation of the variables other than the one at* u *under which the value of* γ *is the value at* u *or its complement; in particular* γ *takes both values.*

PROOF. By induction on γ, every such formula takes both values. Going from u to the root, give the sibling subformula the value 1 at a conjunction or at the antecedent of an implication whose consequent contains u, and the value 0 at a disjunction or at the consequent of an implication whose antecedent contains u. Siblings have disjoint variables, so these requirements are compatible. ⊣

LEMMA 3.9. *If every* C_i *is read-once, then no separation of* $\mathsf{Mon}(\mathcal{C}, r)$ *belongs to* **CL**.

PROOF. Let β be obtained from $\mathsf{Mon}(\mathcal{C}, r)$ by replacing one occurrence of a variable q by a new variable z. Unless $q = D$, we give D the value 0; then β is false as soon as all left members are true and Q or some right member is false. We give such a valuation in each case. By Lemma 3.8, for every occurrence in C_i there are values of the data variables and of z such that C_i is false and the formula obtained from C_i by replacing that occurrence by z is true, and also values with the opposite outcome.

(a) q *is a data variable in* ${H_i}^L$. Let $s = 0$, $x_i = 1$ and $x_j = 0$ for $j \neq i$, and choose the data variables and z so that C_i is false and the modified left copy of C_i is true. All left members are true, and ${H_i}^R$ is false.

(b) q *is a data variable in* ${H_i}^R$. As in (a), but with C_i true and the modified right copy false.

(c) $q = x_i$ *in* ${H_i}^L$. Let $s = 0$, $x_i = 1$, $z = 0$, $x_j = 0$ for $j \neq i$, and make C_i false. Then the left members are true and ${H_i}^R$ is false.

(d) $q = x_i$ *in* ${H_i}^R$. Let all selectors and s be 0, $z = 1$, and make C_i false. Then the modified ${H_i}^R$ is false.

(e) q *is* s *or* x_i *in* ${K_i}^L$. Let all data variables be 1, so that every C_j is true. Let $s = 1$, $x_i = 0$ and $x_j = 1$ for $j \neq i$, and let $z = 0$ if $q = s$ and $z = 1$ if $q = x_i$. The left members are true, and ${K_i}^R = s \to x_i$ is false.

(f) $q = s$ *in* ${K_i}^R$*, or in* Q. Let s, all selectors and r be 0, and $z = 1$. The left members are true, and the modified ${K_i}^R$, resp. Q, is false.

(g) $q = x_i$ *in* ${K_i}^R$*, or* $q = r$ *in* Q. Let all variables other than z and D be 1, and $z = 0$. The left members are true, and the modified ${K_i}^R$, resp. Q, is false.

(h) $q = D$. Let the negative occurrence of D have value 1, the positive one value 0, and all other variables value 1. All antecedents of β are true and its consequent is false.

Every occurrence of a variable in $\mathsf{Mon}(\mathcal{C}, r)$ is covered by one of these cases. ⊣

Example 3.10. For P there are seven local constraints, so Γ has fourteen members, among them $x_{001} \to b \to t_{001}$ and $s \to x_{001}$ (we index the selectors by the occurrences of P). Consider again the separation replacing t_{001} by z, now in the left copy of $x_{001} \to b \to t_{001}$. The valuation with $s = 0$, $x_{001} = 1$, all other selectors 0, $b = z = 1$, $t_{001} = 0$ and $D = 0$ falsifies it: all left members are true, since the switch and the other selectors are off, whereas the right copy $x_{001} \to b \to t_{001}$ is false. No other constraint plays a role.

Splittings that modify several occurrences at once may still survive. For P, replace t_{001} by z in *both* copies of $x_{001} \to b \to t_{001}$. The two copies remain identical, so the query cannot detect the change; and under the switch the modified constraint $b \to z$ is not needed to obtain t_ε, as we saw above. Hence this splitting of $\mathsf{Mon}(\mathcal{C}_P, t_\varepsilon)$ is in **CL**, and so is the abstraction of both copies of $b \to t_{001}$. The next construction prevents modifications of this kind by making all occurrences distinct.

3.3. The tree closure. We say that a formula F is *separation-rigid* if every variable of F has a positive and a negative occurrence and no separation of F belongs to **CL**. We now show that a single further construction turns a separation-rigid formula of **CL** into a minimal one: we rename all occurrences apart and tie the ports of each variable together by a spanning tree. A splitting of a center then cuts an edge of the tree, and a splitting of a port is either refuted by an edge or reduces to a separation of F.

Let F be an implicational formula in which every variable has a positive and a negative occurrence, and let F°, $\mathcal{G}u_F$ be as in Section 3.1. For a center a, let P_a and N_a be the sets of ports of positive and negative occurrences of a, both nonempty. Choose $p_a \in P_a$ and $n_a \in N_a$. The *edge formulas* of a are

$$n \to p_a \quad (n \in N_a) \qquad \text{and} \qquad n_a \to p \quad (p \in P_a \setminus \{p_a\}).$$

Viewed as edges, they form a spanning tree on $P_a \cup N_a$, and every port lies on at least one edge. Let $\mathcal{R}_F$ be the list of all edge formulas and d a new variable. The *tree closure* of F is

$$\mathsf{T}(F) = \mathcal{G}u_F \to (F^\circ \to \mathcal{R}_F \to d) \to d.$$

Since $F^\circ \to \mathcal{R}_F \to d$ is an antecedent of an antecedent, every occurrence in F° has in $\mathsf{T}(F)$ the same polarity as the corresponding occurrence in F.

Example 3.11. In $\mathsf{Mon}(\mathcal{C}_P, t_\varepsilon)$ the variable t_{001} occurs four times: in the left and right copies of $x_{00} \to C_{00}$ and of $x_{001} \to C_{001}$. Let p_1, p_2, p_3, p_4 be the ports of these occurrences, in this order. The first and the last are positive and the other two negative, so the guards are

$$t_{001} \to p_1, \qquad p_2 \to t_{001}, \qquad p_3 \to t_{001}, \qquad t_{001} \to p_4,$$

and, with $p_{t_{001}} = p_1$ and $n_{t_{001}} = p_2$, the edge formulas are $p_2 \to p_1$, $p_3 \to p_1$ and $p_2 \to p_4$. The splitting described at the end of Section 3.2 becomes the splitting

of the center t_{001} that separates the guards of p_3, p_4 from those of p_1, p_2. The edge $p_3 \to p_1$ joins the two classes, and the valuation that gives the value 1 to the center occurrence and the port on the side of p_3, and 0 on the side of p_1, makes all guards true and $p_3 \to p_1$ false. The abstraction of $b \to t_{001}$ disappears as well, since its two copies become $p \to p_3$ and $p' \to p_4$ for distinct ports p, p' of b.

LEMMA 3.12. $\mathbf{IL} \vdash F \to \mathsf{T}(F)$, *and* $\mathbf{IL} \vdash \tau(\mathsf{T}(F)) \to F$ *for the extension* τ *of* τ_F *that maps* d *to* F. *Hence* $\mathbf{IL} + \mathsf{T}(F) = \mathbf{IL} + F$.

PROOF. By Lemma 3.1, $\mathcal{G}u_F \vdash_{\mathbf{IL}} F \to F^\circ$, and every edge formula $n \to p$ of a follows from the guards $n \to a$ and $a \to p$. Hence from F, the guards and $F^\circ \to \mathcal{R}_F \to d$ we obtain d. Conversely, τ maps every guard and every edge formula to a formula $a \to a$, and maps $F^\circ \to \mathcal{R}_F \to d$ to $F \to \tau(\mathcal{R}_F) \to F$. ⊣

LEMMA 3.13. *If* F *is separation-rigid, then no splitting of* $\mathsf{T}(F)$ *belongs to* **CL**.

PROOF. Let β be a splitting of $\mathsf{T}(F)$ that replaces the occurrences in a set Q of a variable q by z. Unless $q = d$, we give d the value 0; then β is false as soon as all guards are true and F° or some edge formula is false.

(a) q *is a center* a. The variable a occurs exactly once in each guard of an occurrence of a and nowhere else, so Q divides the ports of a into two nonempty classes. Since the edge formulas of a form a spanning tree, some edge formula $n \to p$ of a joins the classes. Give the occurrence of a or z in the guard of n the value 1 and the one in the guard of p the value 0, give each port of a the value of the center occurrence in its guard, and give all other variables the value 0. All guards are true and $n \to p$ is false.

(b) q *is a port* p_o *and some occurrence of* p_o *in an edge formula lies on the other side from the one in its guard.* We may assume that the occurrence in the guard is not in Q (otherwise interchange q and z, which yields a renaming of β). If p_o is positive, let $z = 0$ and all other variables 1; then all guards are true, and an edge formula with z as its consequent is false. If p_o is negative, let $z = 1$ and all other variables 0; then an edge formula with z as its antecedent is false.

(c) q *is a port* p_o *and all its occurrences in edge formulas lie on the side of its guard.* Since p_o occurs exactly once in F° and Q is a proper nonempty subset, we may assume that Q is the occurrence in F°. Since F is separation-rigid, the separation of F at o has a falsifying valuation v. Give every center its value under v, every port except z the value of its center, and z the value of the new variable of the separation. All guards and edge formulas are true, and the modified F° has the value of the separation of F under v, that is, 0.

(d) $q = d$. Give the inner occurrence of d the value 1, the outer one 0, and all other variables 1. ⊣

LEMMA 3.14. *Let* $\alpha \in \mathbf{CL}$ *be a formula no splitting of which belongs to* **CL**, *and suppose that an abstraction* $\alpha[Q{:=}z]$, *where* Q *is a set of occurrences of a compound formula* γ, *belongs to* **CL**. *Then there are distinct variables* a, b *such that every occurrence of* a *and of* b *in* α *lies in an occurrence of* $a \to b$.

PROOF. Let ℓ be a leaf of γ, with variable a, and let γ' be γ with a replaced by a new variable a' at ℓ. Substituting γ' for z in $\gamma[Q := z]$ gives a formula

of **CL**, namely the formula obtained from α by replacing a by a' at the set $S_\ell = \{u\ell \mid u \in Q\}$. Since this is not a splitting, $S_\ell = \mathrm{Occ}_\alpha(a)$. If a also occurred at another leaf k of γ, then $S_k = \mathrm{Occ}_\alpha(a)$ as well, although $S_k \cap S_\ell = \emptyset$. Hence γ is read-once and every occurrence of each of its variables lies at the corresponding position of an element of Q. Taking an occurrence in γ of a subformula $a \to b$ with a, b variables, we obtain the claim. ⊣

LEMMA 3.15. *In $\mathsf{T}(F)$ there are no distinct variables a, b such that every occurrence of a and of b lies in an occurrence of $a \to b$.*

PROOF. A subformula of $\mathsf{T}(F)$ of the form $a \to b$ with a, b variables is a guard, or occurs in F°, or is an edge formula; the antecedents of the outer implications of $\mathsf{T}(F)$ and of $F^\circ \to \mathcal{R}_F \to d$ are compound, and d is not the consequent of an implication with a variable antecedent. A guard contains a port, which also occurs in F°. The other two kinds consist of ports, each of which also occurs in its guard, which contains a center. ⊣

THEOREM 3.16. *Let $F \in \mathbf{CL}$ be separation-rigid. Then $\mathsf{T}(F)$ is minimal in $\mathbf{CL}$, $\mathbf{IL} + \mathsf{T}(F) = \mathbf{IL} + F$, $\mathrm{Ord}(\mathsf{T}(F)) \leq \mathrm{Ord}(F) + 2$, and $|\mathsf{T}(F)| = 5|F| - 2|\mathrm{Var}(F)| + 2$.*

PROOF. We have $\mathsf{T}(F) \in \mathbf{CL}$ by Lemma 3.12. No splitting of $\mathsf{T}(F)$ is in **CL** by Lemma 3.13, and hence no abstraction by Lemmas 3.14 and 3.15. By Proposition 2.3, $\mathsf{T}(F)$ is minimal in **CL**. Guards and edge formulas have order 2, which gives the bound on the order. There are $|F|$ guards and $|F| - |\mathrm{Var}(F)|$ edge formulas, which gives the size. ⊣

3.4. Minimal axioms. Let Φ be an implicational formula that is not a variable, and put

$$F_\Phi = \mathsf{Mon}(\mathcal{C}_\Phi, t_\varepsilon), \qquad M_\Phi = \mathsf{T}(F_\Phi).$$

THEOREM 3.17. *Let $\Phi \in \mathbf{CL}$ be an implicational formula that is not a variable. Then M_Φ is minimal in $\mathbf{CL}$, and for the substitution $\sigma_\Phi = \theta' \circ \tau$, which fixes the variables of Φ, where θ' is as in Lemma 3.6 for the substitution of Lemma 3.4 and τ is as in Lemma 3.12,*

$$\mathbf{IL} \vdash \Phi \to M_\Phi, \qquad \mathbf{IL} \vdash \sigma_\Phi(M_\Phi) \to \Phi, \qquad \mathbf{IL} + M_\Phi = \mathbf{IL} + \Phi.$$

Moreover $|M_\Phi| = 102|\Phi| - 2|\mathrm{Var}(\Phi)| - 38$ and $\mathrm{Ord}(M_\Phi) \leq 7$.

PROOF. By Lemmas 3.4 and 3.6, $\mathbf{IL} \vdash \Phi \to F_\Phi$ and $\mathbf{IL} \vdash \theta'(F_\Phi) \to \Phi$; in particular $F_\Phi \in \mathbf{CL}$. The local constraints are read-once and t_ε occurs in C_ε, so F_Φ is separation-rigid by Lemmas 3.7 and 3.9. Theorem 3.16 and Lemma 3.12 give the minimality and the equivalences.

For the size, let $L = |\Phi|$ and $V = |\mathrm{Var}(\Phi)|$. The $2L - 1$ local constraints have total size $2L + 3(L - 1) = 5L - 3$, so the members of Γ have total size $5L - 3 + 3(2L - 1) = 11L - 6$, and $|F_\Phi| = 2(11L - 6) + 4 = 22L - 8$. The variables of F_Φ are those of Φ, the $2L - 1$ node variables, the $2L - 1$ selectors, s and D, that is, $4L + V$ in number. Theorem 3.16 gives the formula. Finally, the local constraints have order at most 3, so $\mathrm{Ord}(F_\Phi) \leq 5$ and $\mathrm{Ord}(M_\Phi) \leq 7$. ⊣

§4. Reduction to order four. We now show that the translation S of Definition 3.2 preserves minimality. Since $\mathsf{S}(\alpha)$ has order at most 4, applying S to the formula M_Φ of Theorem 3.17 proves Theorem 1.1.

4.1. Preservation of minimality.

THEOREM 4.1. *Let L be a logic containing I. If α is minimal in L and $\mathsf{S}(\alpha) \in L$, then $\mathsf{S}(\alpha)$ is minimal in L.*

We use three closure properties of such a logic L: (C1) it is closed under substitution; (C2) if $\gamma_1 \to \cdots \to \gamma_k \to \beta \in L$ and $\gamma_1, \dots, \gamma_k \in L$, then $\beta \in L$; (C3) $\gamma \to \gamma \in L$ for every γ. In the rest of this subsection α is not a variable, $u_1, \dots, u_m$ are its non-leaf occurrences in post-order, and z is a variable not occurring in $\mathsf{S}(\alpha)$. By Proposition 2.3, it suffices to show that if $\mathcal{G}(\mathsf{S}(\alpha)) \cap L \neq \emptyset$, then $\mathcal{G}(\alpha) \cap L \neq \emptyset$.

We name some occurrences in $\mathsf{S}(\alpha)$. For $1 \le i \le m+1$ let $T_i = E_{u_i} \to \cdots \to E_{u_m} \to x_\varepsilon$; it occurs at 1^{i-1}. For $u = u_i$, the formula E_u occurs at $\mathsf{prem}(u) = 1^{i-1}0$; let $\mathsf{body}(u)$ and $\mathsf{def}(u)$ be the occurrences of b_u and x_u in E_u, so that $\mathsf{body}(u) = \mathsf{prem}(u)0$ and $\mathsf{def}(u) = \mathsf{prem}(u)1$ if u is positive, and $\mathsf{body}(u) = \mathsf{prem}(u)1$ and $\mathsf{def}(u) = \mathsf{prem}(u)0$ otherwise. For an occurrence v of α define the occurrence $\mathsf{rep}(v)$ of r_v in $\mathsf{S}(\alpha)$ by $\mathsf{rep}(w0) = \mathsf{body}(w)0$, $\mathsf{rep}(w1) = \mathsf{body}(w)1$ and $\mathsf{rep}(\varepsilon) = 1^m$. Let h denote the height of a formula: $h(p) = 0$ and $h(\gamma \to \delta) = 1 + \max\{h(\gamma), h(\delta)\}$.

LEMMA 4.2.

(a) *Every occurrence in $\mathsf{S}(\alpha)$ is exactly one of 1^{i-1} $(1 \le i \le m)$, $\mathsf{prem}(u)$, $\mathsf{body}(u)$, $\mathsf{def}(u)$ and $\mathsf{rep}(v)$, carrying T_i, E_u, b_u, x_u and r_v, respectively.*

(b) *The variable x_u occurs in $\mathsf{S}(\alpha)$ exactly at $\mathsf{def}(u)$ and $\mathsf{rep}(u)$. A variable c of α occurs exactly at the $\mathsf{rep}(v)$ with v a leaf and $\alpha_v = c$.*

(c) *$h(b_u) = 1$, $h(E_u) = 2$ and $h(T_i) \ge 3$ for $i \le m$. Hence formulas of height 1, 2 and at least 3 occur only at occurrences $\mathsf{body}(u)$, $\mathsf{prem}(u)$ and 1^{i-1}, respectively.*

(d) *If $b_u = b_v$, then $\alpha_u = \alpha_v$. If $E_u = E_v$, then $u = v$. If $T_i = T_j$, then $i = j$.*

PROOF. (a)–(c) follow from Definition 3.2, since the extension variables are distinct and new. (d) If $b_u = b_v$, then $\alpha_u = \theta_\alpha(b_u) = \theta_\alpha(b_v) = \alpha_v$. The formula E_u has a variable as its consequent if and only if u is positive; so $E_u = E_v$ implies that u, v have the same polarity, whence $x_u = x_v$ and $u = v$. If $i < j$, then T_j is a proper subformula of T_i. ⊣

Let $\mathcal{B}(\alpha)$ be the set of $\mathsf{S}(\alpha)[P{:=}z] \in \mathcal{G}(\mathsf{S}(\alpha))$ such that every element of P is of the form $\mathsf{body}(u)$ or $\mathsf{rep}(v)$.

LEMMA 4.3. *Let $\alpha' = \mathsf{S}(\alpha)[P{:=}z] \in \mathcal{B}(\alpha)$ and $Q = \{v \mid \mathsf{rep}(v) \in P\} \cup \{u \mid \mathsf{body}(u) \in P\}$. Then $\beta = \alpha[Q{:=}z] \in \mathcal{G}(\alpha)$, and there are a substitution θ and formulas $\gamma_1, \dots, \gamma_m$ with $\theta\alpha' = (\gamma_1 \to \gamma_1) \to \cdots \to (\gamma_m \to \gamma_m) \to \beta$.*

PROOF. Let P consist of occurrences of δ. First, $\beta \in \mathcal{G}(\alpha)$: clearly $Q \neq \emptyset$ and z does not occur in α. If δ is a variable of α, then by Lemma 4.2(b), Q is a set of leaves carrying δ, not all of them. If $\delta = x_w$, then $P = \{\mathsf{rep}(w)\}$, since $\alpha' \in \mathcal{B}(\alpha)$, so $Q = \{w\}$ with α_w compound. If δ is compound, then $Q = \{u \mid \mathsf{body}(u) \in P\}$ and $\alpha_u = \theta_\alpha(\delta)$ for all $u \in Q$.

No occurrence 1^{i-1}, $\mathsf{prem}(u)$ or $\mathsf{def}(u)$ is in P. Hence for every non-leaf occurrence v of α,

$$\alpha'_{\mathsf{body}(v)} = \begin{cases} z, & \text{if } \mathsf{body}(v) \in P, \\ \alpha'_{\mathsf{rep}(v0)} \to \alpha'_{\mathsf{rep}(v1)}, & \text{otherwise,} \end{cases} \tag{2}$$

where $\alpha'_{\mathsf{rep}(vj)}$ is z or r_{vj}. Let θ fix every variable that is not an extension variable, and let $\theta(x_v) = \theta(\alpha'_{\mathsf{body}(v)})$; by (2) this defines θ by induction on $h(\alpha_v)$.

We claim that $\theta(\alpha'_{\mathsf{rep}(v)}) = \beta_v$ for every occurrence v of α not strictly below an element of Q (then $\mathsf{rep}(v)$ is an occurrence of α', since its proper prefixes are not in P). If $\mathsf{rep}(v) \in P$, both sides are z. Otherwise $\alpha'_{\mathsf{rep}(v)} = r_v$. If v is a leaf, both sides are α_v. If $\mathsf{body}(v) \in P$, both sides are z. Otherwise $v \notin Q$, and by (2) and induction, $\theta(x_v) = \beta_{v0} \to \beta_{v1} = \beta_v$.

Finally $\alpha' = \alpha'_{\mathsf{prem}(u_1)} \to \cdots \to \alpha'_{\mathsf{prem}(u_m)} \to \alpha'_{\mathsf{rep}(\varepsilon)}$, where $\alpha'_{\mathsf{prem}(u_i)}$ is $\alpha'_{\mathsf{body}(u_i)} \to x_{u_i}$ or $x_{u_i} \to \alpha'_{\mathsf{body}(u_i)}$; θ maps it to $\gamma_i \to \gamma_i$ with $\gamma_i = \theta(x_{u_i})$, and maps $\alpha'_{\mathsf{rep}(\varepsilon)}$ to β by the claim. ⊣

LEMMA 4.4. *If $\mathcal{G}(\mathsf{S}(\alpha)) \cap L \neq \emptyset$, then $\mathcal{B}(\alpha) \cap L \neq \emptyset$.*

PROOF. Let $\alpha' = \mathsf{S}(\alpha)[P{:=}z] \in L$ with P a set of occurrences of δ. By (C1) it suffices to find σ with $\sigma\alpha' \in \mathcal{B}(\alpha)$. If δ is a variable of α or $h(\delta) = 1$, then $\alpha' \in \mathcal{B}(\alpha)$ by Lemma 4.2(b),(c). If $\delta = x_u$, then $P = \{\mathsf{rep}(u)\}$ or $P = \{\mathsf{def}(u)\}$; in the second case, interchanging x_u and z gives $\mathsf{S}(\alpha)[\{\mathsf{rep}(u)\}{:=}z]$. If $h(\delta) = 2$, then $P = \{\mathsf{prem}(u)\}$ for some u by Lemma 4.2(c),(d); with $\mathsf{def}(u) = \mathsf{prem}(u)d$, substituting $E_u[\{d\}{:=}z]$ for z gives $\mathsf{S}(\alpha)[\{\mathsf{def}(u)\}{:=}z]$, treated above. If $h(\delta) \geq 3$, then $P = \{1^{i-1}\}$ for some $i \leq m$, and substituting $T_i[\{1^{m-i+1}\}{:=}z]$ for z gives $\mathsf{S}(\alpha)[\{\mathsf{rep}(\varepsilon)\}{:=}z] \in \mathcal{B}(\alpha)$. ⊣

PROOF OF THEOREM 4.1. If α is a variable, there is nothing to prove. Otherwise suppose $\mathcal{G}(\mathsf{S}(\alpha)) \cap L \neq \emptyset$. By Lemma 4.4 there is $\alpha' \in \mathcal{B}(\alpha) \cap L$; take β, θ and γ_i as in Lemma 4.3. By (C1)–(C3), $\beta \in L$, contradicting the minimality of α, since $\beta \in \mathcal{G}(\alpha)$. Hence $\mathcal{G}(\mathsf{S}(\alpha)) \cap L = \emptyset$, and $\mathsf{S}(\alpha)$ is minimal in L by Proposition 2.3. ⊣

4.2. Proof of the main results.

PROPOSITION 4.5. *Every formula of* **CL** *of order at most* 3 *belongs to* **IL**.

PROOF. Such a formula has the form $H_1 \to \cdots \to H_n \to r$, where each H_i is $p_1 \to \cdots \to p_k \to q$ with variables p_j, q, read as the rule "from $p_1, \ldots, p_k$ infer q". Let X be the closure of $\emptyset$ under these rules. The valuation that makes exactly the variables in X true satisfies every H_i, so $r \in X$. Following the construction of X, we derive r from $H_1, \ldots, H_n$ by modus ponens. ⊣

PROOF OF THEOREM 1.1. If Φ is a variable, then $\Phi \notin$ **CL**. Otherwise let $\widehat{M}_\Phi = \mathsf{S}(M_\Phi)$. By Theorem 3.17, M_Φ is minimal in **CL**. By Lemma 3.3, $\mathsf{S}(M_\Phi) \in$ **CL** and $\mathbf{IL} + \widehat{M}_\Phi = \mathbf{IL} + M_\Phi = \mathbf{IL} + \Phi$, and by Theorem 4.1 with $L =$ **CL**, $\widehat{M}_\Phi$ is minimal in **CL**. By (1), $\mathrm{Ord}(\widehat{M}_\Phi) \leq 4$ and $|\widehat{M}_\Phi| = 3|M_\Phi| - 2 = 306|\Phi| - 6|\mathrm{Var}(\Phi)| - 116$. The substitution $\sigma = \sigma_\Phi \circ \theta_{M_\Phi}$ fixes the variables of Φ, and $\mathbf{IL} \vdash \Phi \to \widehat{M}_\Phi$ and $\mathbf{IL} \vdash \sigma(\widehat{M}_\Phi) \to \Phi$ by Lemmas 3.12 and 3.3; then also

$\mathbf{IL} \vdash \Phi \to \sigma(\widehat{M}_\Phi)$, as $\sigma(\Phi) = \Phi$. All constructions are syntax-directed, so $\widehat{M}_\Phi$ is computed in linear time. ⊣

Proof of Corollary 1.2. Let L be an implicational intermediate logic. Then $L = \mathbf{IL} + L$, and replacing each $\Phi \in L$ that is not in **IL** by $\widehat{M}_\Phi$ gives the required Γ (if $L = \mathbf{IL}$, take $\Gamma = \{I\}$). If $L \neq \mathbf{IL}$, any such Γ contains a formula outside **IL**, which has order at least 4 by Proposition 4.5. ⊣

Remark 4.6. The axioms of the base logic are themselves minimal in **CL**: K and S are minimal in **CL** [NM21, Example 2.3], and so are B, C and I [NM21, Sect. 5.3]; moreover $\mathbf{IL} = \mathbf{BCK} + S$. Hence every implicational intermediate logic is $(\{K, S\} \cup \Gamma)^* = \mathbf{BCK} + (\{S\} \cup \Gamma)$ for a set Γ of formulas of order at most 4 minimal in **CL**.

Example 4.7. For Peirce's law, $|M_P| = 366$ and $|\widehat{M}_P| = 1096$. The construction is uniform and far from optimal in size: for the axiom G' of **LC** of Nakamura and Matsuda, $\mathsf{S}(G')$ is already minimal in **CL** by Theorem 4.1, has order 4, and has size 67.

§5. The full language. In this section formulas are built from variables and $\bot$ by $\wedge$, $\vee$ and $\to$. For a nonempty list $\Delta = (\delta_1, \dots, \delta_k)$ we write $\bigwedge \Delta$ for $\delta_1 \wedge (\delta_2 \wedge \cdots \wedge \delta_k)$. Elementary generalizations now include abstractions of sets of occurrences of $\bot$, and Proposition 2.3 applies as stated.

A formula of **CPC** with a positive occurrence of $\bot$ is never minimal in **CPC**: replacing that occurrence by a new variable z gives a formula implied by the original one in **IPC**, by ex falso and monotonicity, and this formula is an abstraction. Our construction therefore keeps $\bot$ only in negative positions.

5.1. Local constraints and the monitor. Let $\Phi \in \mathbf{CPC}$; then Φ is neither a variable nor $\bot$. The local constraints are defined as in Section 3.1: variables are renamed apart with their guards, every non-leaf occurrence receives an extension variable, and so does every occurrence of $\bot$. Writing t_v for the node variable of v, with $u = v0$ and $w = v1$, the constraint C_v is

Φ_v	v positive	v negative
a variable p	$p \to t_v$	$t_v \to p$
$\Phi_u \circ \Phi_w$, $\circ \in \{\wedge, \vee, \to\}$	$(t_u \circ t_w) \to t_v$	$t_v \to (t_u \circ t_w)$
$\bot$	(none)	$t_v \to \bot$

No constraint is needed for a positive $\bot$, since $\bot \to t_v$ is provable. Let W be the set of occurrences that have a constraint, and $W_\bot \subseteq W$ the set of negative occurrences of $\bot$. Every local constraint has order at most 3, and every one not of the form $t_v \to \bot$ is read-once without $\bot$. For a valuation η of the variables of Φ, the *standard valuation* ν_η extends η by $\nu_\eta(t_v) = \eta(\Phi_v)$.

Lemma 5.1. *The local constraints derive $\Phi \to t_\varepsilon$ in* **IPC**. *The substitution $t_v \mapsto \Phi_v$, fixing the variables of Φ, maps every local constraint to a formula $\gamma \to \gamma$; consequently every local constraint is true under every standard valuation.*

Proof. As in Lemmas 3.1 and 3.3, using the monotonicity of $\wedge$ and $\vee$; a positive $\bot$ is handled by $\bot \to t_v$. ⊣

The monitor is modified so as not to copy $\bot$ into a positive position. We take selectors x_v ($v \in W$), a switch s, and new variables e_v ($v \in W_\bot$), and put $H_v = x_v \to C_v$, $K_v = s \to x_v$, $Q = s \to t_\varepsilon$ as before, and

$$\widehat{C_v} = t_v \to e_v \ (v \in W_\bot), \qquad \widehat{C_v} = C_v \ (v \notin W_\bot), \qquad \widehat{H_v} = x_v \to \widehat{C_v}.$$

With Γ the list of the H_v, K_v and $\widehat{\Gamma}$ the list of the $\widehat{H_v}, K_v$ ($v \in W$), let

$$F_\Phi = \bigwedge \Gamma \to \Big(Q \wedge \bigwedge \widehat{\Gamma}\Big).$$

The conjunction replaces the continuation $(Q \to \Gamma \to D) \to D$ of Section 3.2, which lowers the order by one. The left and right members are those in Γ and in $\widehat{\Gamma}$. Each e_v occurs exactly once, and $\bot$ occurs only in left members, negatively.

Lemma 5.2.

(a) $\mathbf{IPC} \vdash \Phi \to F_\Phi$, *and* $\mathbf{IPC} \vdash \rho_\Phi(F_\Phi) \to \Phi$ *for the substitution* ρ_Φ *that fixes the variables of* Φ *and maps* $t_v \mapsto \Phi_v$, $s, x_v \mapsto \top$ *and* $e_v \mapsto \bot$.

(b) *Every variable of* F_Φ *other than the* e_v *has a positive and a negative occurrence.*

(c) *No separation of* F_Φ *belongs to* **CPC**.

(d) *No abstraction of a nonempty set of occurrences of* $\bot$ *in* F_Φ *belongs to* **CPC**.

Proof. (a) As in Lemma 3.6, using Lemma 5.1; the right member $x_v \to t_v \to e_v$ follows from $x_v \to t_v \to \bot$ by ex falso. Under ρ_Φ the members of Γ become theorems, and the first conjunct of the consequent becomes $\top \to \Phi$.

(b) As in Lemma 3.7; a node variable of a positive $\bot$ occurs in the constraint of its parent.

(c) We follow the proof of Lemma 3.9, with every $e_v = 0$, so that $\widehat{C_v}$ and C_v have the same value. Now β is false as soon as all left members are true and Q or some right member is false, and case (h) disappears. In cases (a)–(d) we use Lemma 3.8, and for $C_v = t_v \to \bot$ the fact that its value is the complement of that of t_v. In cases (e) and (g), instead of giving all data variables the value 1, we give them the values of a standard valuation, under which every local constraint is true by Lemma 5.1.

(d) Let the chosen set contain the $\bot$ of ${H_v}^L = x_v \to t_v \to \bot$ and let it be replaced by z. Let $z = 1$, $s = 0$, $x_v = t_v = 1$, $x_w = 0$ for $w \neq v$, and $e_v = 0$. All left members are true, whereas $\widehat{H_v} = x_v \to t_v \to e_v$ is false. ⊣

5.2. The tree closure. Let F be a formula in which every variable that occurs at least twice has a positive and a negative occurrence. We rename apart the occurrences of these variables only, leaving $\bot$ and the variables that occur once unchanged in F°, and define guards and edge formulas as in Section 3.3. In this section the tree closure of F is

$$\mathsf{T}(F) = \bigwedge \mathcal{G}u_F \to \Big(F^\circ \wedge \bigwedge \mathcal{R}_F\Big).$$

Theorem 5.3. *Let* $F \in$ **CPC** *be as above, and suppose that no separation of* F *and no abstraction of a nonempty set of occurrences of* $\bot$ *in* F *belongs to* **CPC**, *and that every subformula* $a \circ b$ *of* F *with* a, b *variables and* $\circ \in \{\wedge, \vee, \to\}$

contains a variable that occurs at least twice in F. Then $\mathsf{T}(F)$ is minimal in **CPC**, $\mathbf{IPC} + \mathsf{T}(F) = \mathbf{IPC} + F$, *and* $\mathrm{Ord}(\mathsf{T}(F)) \leq \max\{3, \mathrm{Ord}(F)\}$.

Proof. The equivalence is proved as in Lemma 3.12, using the monotonicity of $\wedge$ and $\vee$.

Splittings. A splitting replaces occurrences of a center or of a port. With "F° or some edge formula is false" as the condition for falsity, cases (a)–(c) of the proof of Lemma 3.13 apply verbatim; the values of the variables that occur once are taken from the separation in case (c) and are arbitrary otherwise.

Abstractions of $\bot$. The occurrences of $\bot$ in $\mathsf{T}(F)$ are those in F°. Given an abstraction, take a falsifying valuation of the corresponding abstraction of F, give every port the value of its center, and keep the values of the other variables. All guards and edge formulas are true, and the modified F° is false.

Abstractions of compound formulas. The proof of Lemma 3.14 applies, with one addition: if the abstracted formula γ contained $\bot$ at a leaf ℓ, then substituting for z the formula γ with that $\bot$ replaced by a new variable would give an abstraction of $\bot$ in **CPC**. Hence there are distinct variables a, b and $\circ \in \{\wedge, \vee, \to\}$ such that all occurrences of a and b lie in occurrences of $a \circ b$. As in Lemma 3.15, this is impossible: such a subformula is a guard, or an edge formula, or lies in F° and then contains a port by the last hypothesis; and every port also occurs in its guard. The conjunctions joining the members, and the main implication, connect compound formulas.

By Proposition 2.3, $\mathsf{T}(F)$ is minimal in **CPC**. Guards and edge formulas have order 2, so $\mathrm{Ord}(\mathsf{T}(F)) \leq \max\{3, \mathrm{Ord}(F)\}$. ⊣

Let $M_\Phi = \mathsf{T}(F_\Phi)$.

Theorem 5.4. *For every $\Phi \in$* **CPC**, *the formula M_Φ is minimal in* **CPC**, $\mathrm{Ord}(M_\Phi) \leq 4$, $\mathbf{IPC} \vdash \Phi \to M_\Phi$ *and* $\mathbf{IPC} \vdash \sigma_\Phi(M_\Phi) \to \Phi$ *for a substitution σ_Φ fixing the variables of Φ; hence* $\mathbf{IPC} + M_\Phi = \mathbf{IPC} + \Phi$. *Moreover, if Φ has B_+ positive and B_- negative occurrences of $\bot$, then* $|M_\Phi| = 102|\Phi| - 48B_+ - 8B_- - 2|\mathrm{Var}(\Phi)| - 48$.

Proof. By Lemma 5.2 and Theorem 5.3, with $\sigma_\Phi = \rho_\Phi \circ \tau$; the last hypothesis of Theorem 5.3 holds, since the only variables occurring once in F_Φ are the e_v, which occur in $t_v \to e_v$. Every local constraint has order at most 3, so $\bigwedge \Gamma$ has order at most 3 and $\mathrm{Ord}(F_\Phi) \leq 4$. For the size, let $L = |\Phi|$ and $V = |\mathrm{Var}(\Phi)|$. There are $n = 2L - 1 - B_+$ local constraints, of total size $5L - 2B_+ - 3$, so $f := |F_\Phi| = 2(11L - 5B_+ - 6) + 2$. The number of ports is $m = f - 2B_-$ and the number of centers is $w = V + (2L-1) + n + 1$. Hence $|M_\Phi| = 2m + f + 2(m - w)$, which gives the formula. ⊣

Proof of Theorem 1.3. Take $M = M_\Phi$ of Theorem 5.4. Every intermediate logic L equals $\mathbf{IPC} + L$, and replacing each axiom $\Phi \in L$ by M_Φ gives an axiomatization by formulas minimal in **CPC**. ⊣

[Plot: Hilbert-style axiomatization of **IPC** by formulas minimal in **CPC** (check each axiom), for the analogue of Remark 4.6.]

5.3. Lower orders. In the full language the bound 4 is not optimal for every logic.

EXAMPLE 5.5. The formula $p \vee (p \to q)$ has order 2, is minimal in **CPC**, and axiomatizes **CPC** over **IPC**: substituting $\bot$ for q gives $p \vee \neg p$, and conversely $p \vee \neg p$ implies $p \vee (p \to q)$ by ex falso. Its elementary generalizations are refuted as follows: the two splittings of p by giving the first occurrence of p the value 0, the second the value 1, and $q = 0$; the abstraction $p \vee z$ by $p = z = 0$; and z by $z = 0$.

EXAMPLE 5.6. The formula $(p \to q) \vee (\neg p \to r)$ has order 3, is minimal in **CPC**, and axiomatizes Jankov's logic **KC** $=$ **IPC** $+ (\neg p \vee \neg\neg p)$ [Jan68b]: substituting $\bot$ for q and r gives $\neg p \vee \neg\neg p$, and conversely each disjunct of $\neg p \vee \neg\neg p$ implies a disjunct of the formula by ex falso. The elementary generalizations are refuted as follows: the two splittings of p by giving the first occurrence of p the value 1, the second the value 0, and $q = r = 0$; abstracting $\bot$ by $p = z = 1, q = r = 0$; abstracting $p \to q$ by $z = p = r = 0$; abstracting $\neg p$ by $p = z = 1, q = r = 0$; abstracting $\neg p \to r$ by $p = 1, q = z = 0$; and abstracting the whole formula.

REMARK 5.7. $\bot$ cannot be eliminated: **KC** and **IPC** have the same $\bot$-free theorems [Jan68b], so no $\bot$-free formula Ψ satisfies **IPC** $+ \Psi =$ **KC**. In particular, Theorem 1.1 does not yield Theorem 1.3.

§6. Principal types. Let Γ be a set of implicational formulas and $L =$ **IL**$+\Gamma$. Consider the λ-*terms* built from variables and constants c_γ ($\gamma \in \Gamma$) by abstraction and application, typed by the rules of the simply typed λ-calculus together with $c_\gamma : \sigma\gamma$ for every substitution σ; that is, each constant has the type scheme γ, and different occurrences of a constant may receive different instances. We write $\vdash t : \alpha$ if the closed term t has type α. By the Curry–Howard correspondence, $\alpha \in L$ if and only if $\vdash t : \alpha$ for some closed term t: the simply typed λ-calculus inhabits exactly the formulas of **IL**, constants give the instances of Γ, application is modus ponens, and typings are closed under substitution. For **CL** one may take $\Gamma = \{P\}$, so that c_P is a `call/cc` operator.

As for the simply typed λ-calculus [Hin69], every typable term t has a *principal type* $\mathrm{pt}(t)$: the types of t are exactly the substitution instances of $\mathrm{pt}(t)$. The usual unification argument applies, a constant contributing a fresh copy of its type scheme at each occurrence.

[Plot: Give a reference for principal types with constants of schematic type.]

PROPOSITION 6.1. *Let $\alpha \in L$. Then α is minimal in L if and only if α is, up to renaming, the principal type of every closed term t with $\vdash t : \alpha$.*

PROOF. Suppose α is minimal in L and $\vdash t : \alpha$. Then $\mathrm{pt}(t) \leqslant_{\mathrm{sub}} \alpha$ and $\mathrm{pt}(t) \in L$, so $\alpha \leqslant_{\mathrm{sub}} \mathrm{pt}(t)$. Conversely, suppose $\beta \in L$ and $\beta <_{\mathrm{sub}} \alpha$, and let $\vdash s : \beta$. Then $\vdash s : \alpha$, and $\mathrm{pt}(s) \leqslant_{\mathrm{sub}} \beta <_{\mathrm{sub}} \alpha$, so α is not the principal type of s. ⊣

Since $\mathrm{Min}(\mathbf{CL}) \cap L \subseteq \mathrm{Min}(L)$, Theorem 1.1 gives the following.

COROLLARY 6.2. *Every implicational intermediate logic L is* **IL** $+ \Gamma$ *for a set Γ of formulas of order at most 4 such that each $\gamma \in \Gamma$ is the principal type of every closed term of type γ, in the calculus for L as well as in the calculus for* **CL**.

By Remark 2.2, the version of Corollary 6.2 that refers only to the calculus for L is trivial; the point is the calculus for **CL**.

Two facts put the corollary in perspective. On the one hand, Hindley's theorem [Hin69] says that every inhabited simple type is the principal type of some closed term, so being principal for *some* inhabitant is no restriction. On the other hand, being principal for *all* inhabitants is a strong restriction in substructural settings: the formulas minimal in **BCK** are exactly the principal types of $\beta\eta$-normal **BCK**-λ-terms, and each of them has a unique $\beta\eta$-normal inhabitant [Hir92, Hir93, Tat06]; see also [Hir91, KH93]. In **IL** and **CL**, uniqueness fails already for minimal formulas of depth 3 [Tat93]. Corollary 6.2 shows that, in the intermediate setting, the remaining requirement of principality for all classical inhabitants is no restriction on the logics that can be axiomatized.

[Plot: Check the exact statements in [Hin69] (normal forms; combinatory logic vs. λ-calculus) and Tatsuta's notion of depth.]

§7. Concluding remarks. Every implicational intermediate logic, and every intermediate logic in the full language, is axiomatizable by formulas minimal in classical logic, and in fact every single axiom can be replaced by such a formula of order at most 4. We list some questions that remain.

1. Komori's original question concerns extensions of **BCK**. By Remark 4.6, every logic between **IL** and **CL** is **BCK** extended by formulas minimal in **CL**. Our construction uses contraction (the members of Γ are used twice in Lemma 3.6, and guards are shared by several edge formulas), so it does not apply to logics between **BCK** and **IL**. Is every extension of **BCK** contained in **CL** axiomatizable over **BCK** by formulas minimal in **CL**?
2. In the full language, what is the least order of a formula minimal in **CPC** axiomatizing a given intermediate logic? Examples 5.5 and 5.6 show that it can be 2 or 3.
3. What is the least size of a formula minimal in **CL** axiomatizing a given logic, compared with the size of a given axiom?
4. Can an independent axiom system consisting of formulas minimal in **CL** always be chosen [NM21, Open Problem 3]?

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[Plot: Required by the JSL guidelines: acknowledge the use of AI tools in the research and in the preparation of the manuscript, for example: "The authors used Claude (Anthropic) to assist with ... ; all results and the final text were checked by the authors." Adjust to the actual use.]

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